

Windows 11 Optimizer

One menu for every tweak, with backups and one-by-one revert

Read this first

- **Use at your own risk.** These scripts edit the Windows registry and power settings. Use them only on a PC you own. Always keep the restore point the script creates.
- Built for **Windows 11 desktops used for gaming**. On a laptop expect more heat and faster battery drain (processor minimum 100%, power throttling off).
- The gains are small and situational. GPU drivers, in-game settings, XMP/EXPO and cooling matter far more. See the honest notes on each tweak below.
- Some antivirus programs flag `.bat` files that edit the registry. Every file is plain text: open and read it before you run it.

Requirements

- Windows 11 and an administrator account (right-click > *Run as administrator*).
- Keep every folder next to the scripts. Do not move a `.bat` file out of the folder.
- System Protection (System Restore) turned on for the system drive, so the restore point can be created.

Folder layout

File / folder	Purpose
1_Start_Here - PC_Optimizer_Master (Run as Administrator).bat	Main menu. Start here; everything is reachable from it.
2_Revert_Everything - System_Restore (Run as Administrator).bat	Roll the whole PC back with System Restore (serious problems only).
3_Check_Status (Run as Administrator).bat	Check which tweaks are still applied, for example after a Windows update.
4_Network_Test.bat	Read-only ping, jitter and packet-loss test. No administrator rights needed.
5_System_Report.bat	Read-only report: what uses your CPU and RAM, what starts with Windows, disk space, GPU driver age. No administrator rights needed.
tools/	PowerShell helpers used by the menu, the status check and the two tools above. Do not delete.
reg/1_Recommended	Files 01-08: safe for most PCs.
reg/2_Optional	Files 09-13 and 15: the script asks about each one. (16 is a power setting, see the optional tweaks.)
reg/3_Power	File 14: unlocks hidden CPU options in Power Options.
reg_undo/	One undo file per tweak (Windows defaults), used by menu 7.
repair-tools/	Four stand-alone repair tools. Unrelated to the Master menu.
Backup/ (created when you run)	Copy of every key before it is changed, in a dated folder.
OptimizerLog_*.txt, CheckReport_*.txt (created when you run)	Log of every command, and the result of each status check.

Quick start

Three steps

1 Open 1_Start_Here - PC_Optimizer_Master

- Right-click > *Run as administrator*.

2 Type 5 and press Enter

- The script creates a restore point, then runs menus 1, 2, 3 and 4 in order.
- Menu 2 asks about each optional tweak. Answer for your own use (see the notes on files 09-13).
- A restore point normally takes 1-5 minutes. After 10 minutes the script stops waiting and asks whether to continue.

3 Restart once

- Needed for GPU scheduling, Memory Integrity and other system-level values.
- Then work through the manual Windows settings later in this guide.
- After a big Windows update, or every 1-2 months, run **3_Check_Status**.

Menu of PC_Optimizer_Master

Key	Name	What it does
1	Recommended tweaks	Imports files 01-08.
2	Optional tweaks	Asks about 09-13, 15 and 16 one by one. File 13 needs you to type YES.
3	Base system setup	Power plan, services, SSD/HDD check before touching Prefetcher, TCP settings on every network adapter, cleanup of obsolete values, temp files.
4	Power settings	Imports file 14, then sets processor minimum 100%, core parking off, USB selective suspend off, PCIe power saving off (plugged-in profile only).
5	RUN EVERYTHING	Restore point + 1 + 2 + 3 + 4. Recommended the first time.
6	Restore point only	Creates the restore point and nothing else.
7	Revert ONE tweak	Back to the Windows default: 1-14, T = network values, P = power plans.
8	Check disk type	Reports SSD or HDD. Changes nothing.
9	Power Options	Opens the Power Options window to inspect values.
C	Check status	Same as 3_Check_Status.
N	Network test	Same as 4_Network_Test. Changes nothing.
R	System report	Same as 5_System_Report. Changes nothing.
0	Exit	Exits and shows where the log was saved.

If something goes wrong

Three levels of undo: pick the smallest one that fixes it

A Revert one tweak: menu 7

- Example: the mouse feels odd after file 10. Type 7, then 10.
- Restores the Windows default from `reg_undo`. T removes the network values, P resets all power plans.
- The script remembers the revert; the status check shows it as SKIPPED and will not offer to apply it again.

B Restore your exact previous values: Backup folder

- Every run exports the old keys to `Backup\date-time`.
- Double-click a file there to restore the value your PC really had (more precise than the Windows default).

C Roll back the whole PC: 2_Revert_Everything

- Uses System Restore to return to the `Before_PC_Optimizer` restore point.
- The PC restarts by itself, and programs installed after that point are removed. Serious problems only.

Status check

3_Check_Status, or menu C. Use it after a Windows update or every 1-2 months

1 Run 3_Check_Status as administrator

- It checks every item and prints each line with the current value against the expected one.

2 Answer the fix questions

- Recommended items: Y = re-apply everything missing, S = choose one by one, N = change nothing.
- Optional items are always asked one by one; file 13 still needs YES.
- The old values are backed up to `Backup\date-time_check` before anything is re-applied.

3 Automatic re-check

- After fixing, the script checks the fixed items again to confirm. Everything is saved to `CheckReport_date-time.txt`. Restart once.

Status	Meaning	What the script does
OK	The value is still as the script set it.	Nothing.
CHANGED	A value was changed, for example by a Windows update.	Can re-apply it.
MISSING	The value is not there at all.	Can re-apply it.
NOT CHOSEN	An optional tweak (09-13) you never answered.	Asks: Y apply, N not now, X never ask again.
SKIPPED	You skipped or reverted it.	Does not ask or change it.
UNKNOWN	Cannot be read, or this PC does not have the option.	Information only.

What it checks: all 14 .reg files (read from the same files that are applied), the power plan (Ultimate Performance, or High Performance if Ultimate is unavailable), the four power values, Prefetcher (SSD must be 0, HDD must be 3), TcpAckFrequency and TCPNoDelay on every adapter, and the DiagTrack / dmwappushservice services. Optional 15 and 16 are checked too (16: boost mode 2 and preference 0 while plugged in).

Limits: it tells you a value is still there, not that it still has an effect (item 08 can show OK on a GPU that does not support GPU scheduling). Values under GameConfigStore (01, 09) are rewritten by Windows and the Xbox Game Bar, so CHANGED there is common. Adding a new VPN or Wi-Fi adapter makes Network show CHANGED until it is configured.

.reg files, group 1: Recommended (01-08)

Menu 1. Meant to be safe for most PCs

01 GameBar_DVR_Off

- Turns off Xbox Game Bar background recording (Game DVR), at user and policy level. Game Mode is kept on (v3 turned it off by mistake).

Helps: gives back CPU, GPU and disk time used for constant recording. No official Microsoft documentation for these values.

02 Power_Throttling_Off

- PowerThrottlingOff = 1 stops Windows slowing background programs. Energy estimation off. Fast Startup off, so Shutdown is a real clean restart.

Helps: Discord or OBS in the background are not slowed down. Cost: slightly longer boot, and more power use on laptops.

03 Gaming_Priority_MMCSS

- NetworkThrottlingIndex = ffffffff removes the network limit for multimedia tasks.
- SystemResponsiveness = 10, the lowest value that takes effect. Microsoft documents that values below 10 are raised to the default of 20, so a value of 0 does nothing.
- Tasks\Games: GPU Priority 8, Priority 6, Scheduling Category High, SFIO Priority High.

Helps: queues work so the game gets CPU, GPU and disk first.

04 CPU_Foreground_Priority

- Win32PrioritySeparation = 0x26: the active window gets longer, higher-priority CPU time slices (default is 0x2).

Helps: the game in front gets more CPU than background programs.

05 Telemetry_Off

- AllowTelemetry = 0, CEIPEnable = 0. On Home and Pro, Windows uses the lowest level it allows; true 0 only applies to Enterprise.

Helps: less background data sent. Tiny effect on FPS; mostly privacy.

06 System_Responsiveness

- Menus appear instantly, frozen apps are closed faster, lighter animations and visual effects.
- NTFS last-access time stamps off; the classic Copy To / Move To right-click entries restored.

Helps: Windows feels quicker, fewer needless disk writes.

07 Delivery_Optimization_Off

- DODownloadMode = 0.

Helps: Windows Update does not upload updates to other PCs while you play online.

08 GPU_Scheduling_On

- HwSchMode = 2 turns on hardware-accelerated GPU scheduling. Needs a restart and a supported GPU and driver; unsupported systems ignore it.

Helps: slightly lower CPU-to-GPU latency. Microsoft says most users should not expect a big change.

Optional tweaks (09-13, 15, 16)

The script asks about each one. Answer for how you actually play

09 Fullscreen_Exclusive

- Forces true Fullscreen Exclusive instead of letting Windows convert games to borderless.
- Answer Y only if fullscreen games stutter or hang on alt-tab. Some newer games work better the default way.

Helps: lower input lag in older games that support Fullscreen Exclusive.

10 Mouse_No_Acceleration

- Turns off Enhance Pointer Precision so the mouse moves 1:1.

Helps: consistent aim in shooters.

11 Keyboard_Fast_Repeat

- KeyboardDelay = 0 (shortest delay before repeat). Flags = 122 stops the Filter Keys pop-up when Shift is held for 8 seconds in a game.

Helps: faster key response and no interrupting pop-up.

12 Ndu_Memory_Fix

- Disables the Ndu service (a memory leak reported on some PCs) and sets DisablePagingExecutive = 1.
- Side effects: Task Manager stops showing per-app network history. Keeping the kernel in RAM costs memory, so skip it on PCs with 8 GB or less.

Helps: only if your RAM use keeps creeping up. The Ndu leak comes from user reports, not Microsoft documentation.

13 Memory_Integrity_Off

- Turns off Memory Integrity (HVCI). You must type YES, and restart.
- Microsoft confirms it can affect gaming performance on some PCs (no figure given). With it off, Windows protects less against kernel-level malware.
- Skip it on a work PC or one holding important data. Undo with menu 7, then 13.

Helps: the only tweak here that can raise FPS noticeably, at a security cost.

15 Timer_Resolution

- Sets GlobalTimerResolutionRequests = 1. Since Windows 10 2004 and Windows 11, a game's timer-resolution request only applies to that game. This value lets such a request apply system-wide again.
- Can steady frame pacing in some games. Microsoft notes that a higher timer resolution uses more power and can keep the CPU from saving power. Needs a restart. Undo: menu 7, then 15.

Helps: frame pacing in some games, effect varies. The value comes from the tweaking community; Microsoft does not document it.

16 CPU_Boost_Aggressive (power setting, no .reg file)

- Plugged in only: processor performance boost mode = Aggressive (2) and energy performance preference = 0 (favour performance), on the active power plan.
- Your earlier values are saved first, so menu 7, then 16 restores exactly those. The Ultimate Performance plan already uses these values on most PCs, so this mainly matters on High Performance or Balanced. Expect more heat, especially on laptops.

Helps: 1% lows in games limited by the CPU. Skipped automatically on PCs that do not offer these settings.

Group 3: Power (14) and automatic settings

Menu 4 does the whole job, no manual setup

14 Unlock_CPU_Power_Settings

- Makes visible the CPU options Windows hides: core parking, maximum frequency, boost mode, energy performance preference.
- Merges files 11 and 12 of v3 and drops entries that are normally visible.

P What menu 4 then sets with powercfg

- Minimum processor state = 100%. Core parking minimum = 100% (cores never sleep).
- USB selective suspend = off. PCIe Link State Power Management = off.
- Plugged-in values only; the battery profile on a laptop is not touched.

Helps: CPU and GPU do not switch to power-saving states mid-game.

What menu 3 (Base setup) does

The parts that are not plain .reg files

Item	What it does	Note
Power plan	Uses Ultimate Performance; High Performance if that is missing; leaves Balanced if neither exists.	The plan GUID is remembered, so repeat runs do not create duplicates (v3 did).
Services	Windows Search and Print Spooler on; DiagTrack and dmwappushservice off.	"already running" in the log is normal.
Prefetcher	Checks the disk first: SSD off, HDD on, unknown type left alone.	Stops you disabling it on an HDD, which would slow it down.
Network	Sets TcpAckFrequency and TCPNoDelay on every network adapter.	v3 set these globally, where Windows ignores them. Only affects games using TCP. Microsoft advises testing before changing these.
Old values	Removes MaxUserPort, TcpTimedWaitDelay and misplaced TCP values from v3.	None of them matter for gaming.
Temp files	Deletes the user's temporary files.	Files in use are skipped safely.

Windows settings to change by hand

Things a script cannot do for you

Already automatic in v4, no need to do by hand: GPU scheduling (08), Delivery Optimization (07), USB selective suspend and PCIe power saving (menu 4), Game Bar DVR (01), Game Mode kept on (01), Enhance Pointer Precision (10, if you answered Y).

Clear effect: do these	Where	Why
GPU power = Maximum Performance	NVIDIA Control Panel / AMD Software	Stops the card down-clocking under uneven load.
Network adapter power saving off	Device Manager > adapter > Power Management	Avoids ping spikes.
Add game folders to Defender exclusions	Virus & threat protection > Exclusions	Faster level loading. Does not turn Defender off.
Turn off GPU-software overlays	NVIDIA App / AMD Software > Settings	Less background CPU/GPU load.
Disable audio enhancements	Control Panel > Sound > Enhancements	Lower audio latency.
Cap FPS 3-4 below the refresh rate	In-game limiter, RTSS or NVIDIA Control Panel	Steadier frame times, lower input lag.

NVIDIA Reflex / Low Latency Mode	In-game setting, or NVIDIA Control Panel	Lower input lag when the GPU is the limit.
Shader Cache Size = Unlimited (NVIDIA)	NVIDIA Control Panel > Manage 3D settings	Fewer stutters from shaders being rebuilt.
Use a network cable, not Wi-Fi	Router to PC	The biggest fix for ping spikes and jitter.

Situational or small effect	Where	Why
Focus Assist while gaming	Settings > System > Notifications	No notifications stealing focus.
Xbox Game Bar fully off (if you do not use Win+G)	Settings > Gaming > Xbox Game Bar	One less background process.
Disable fullscreen optimizations per game	Right-click .exe > Properties > Compatibility	For older games with problems.
GPU vendor telemetry services off	services.msc	Less background work.
Move Automatic Maintenance	Control Panel > Security and Maintenance	No scans during games.
Transparency and HDR off if unused	Settings > Personalization / Display	Small saving.
Exclude game folders from Search indexing	Settings > Privacy & security > Searching Windows	Less disk I/O.
Stop unused background apps	Settings > Apps > Installed apps	Frees a little RAM.
Fixed-size page file	System Properties > Advanced > Virtual Memory	More stable over time.

Make sure of: GPU drivers installed from the vendor directly, XMP/EXPO enabled in the BIOS, Variable Refresh Rate on if your monitor supports it, and games set to use the "High performance" GPU on dual-GPU laptops.

Diagnostic tools

Read-only: they change nothing and need no administrator rights. Menu N and R run the same tools

4 4_Network_Test.bat

- Sends 30 pings each to your router, 1.1.1.1, 8.8.8.8 and, if you type one, a game server (these are ordinary ping packets). It shows average ping, jitter and packet loss, grades each line Good / Fair / Poor and says where the problem most likely is.
- Jitter** is how much the delay jumps around. It is what you feel as rubber-banding, and it matters more than a slightly higher average ping.
- The result is saved as `NetworkTest_date-time.txt` next to the scripts.

What you see	Most likely cause	What to try
Router line is Fair or Poor	Wi-Fi signal, a bad cable, or other devices using the network.	Use a cable, move closer, pause other downloads and streams.
Router is Good, internet lines are Poor	Your internet provider or the line.	Restart the router; if it stays, contact the provider.
Router and internet Good, only the game server is bad	The route to that server, or the server itself.	Choose another server or region. Nothing on your PC fixes this.
Game server shows "No reply"	Many servers block ping.	Not a problem by itself.

5 5_System_Report.bat

- Lists Windows, CPU, RAM, graphics card and driver age, disks and free space, the power plan, the programs using the most CPU and RAM (browsers with many processes are added together), and the programs that start with Windows.
- Ends with plain hints: long uptime, RAM nearly full, an old GPU driver, a drive with less than 15% free, a failing disk, too many startup programs.
- Saved as `SystemReport_date-time.txt`. Your Windows user name is hidden in the file, but check it before you share it. Program names and paths are still listed.

What really helps steady FPS and low ping

- Cap the frame rate 3-4 below your monitor's refresh rate (in the game, RTSS or the NVIDIA Control Panel). It gives steadier frame times than an uncapped game that pins the GPU.
- Turn on NVIDIA Reflex (or Low Latency Mode) if your game offers it.
- Use a network cable. Wi-Fi jitter is the most common cause of bad ping.
- Keep GPU drivers current, keep the PC cool and close heavy programs before you play.
- None of the registry tweaks in this project lowers ping by a noticeable amount. They are small, situational gains.

Folder: repair-tools

Targeted repair tools. Not needed on every run. Right-click > Run as administrator

A Deep_Clean_Junk_Files.bat

- Runs Disk Cleanup for every category, plus temp files. Reports free space before and after, and warns if less than 15% is left.

Use when: disk space is low, or every 1-2 months.

B Network_Reset_Fix_Internet.bat

- Resets Winsock, IPv4 and IPv6, renews the IP address, flushes DNS and ARP, then pings to test.

Use when: the connection drops, will not connect, or is oddly slow.

C Bluetooth_Fix.bat

- Re-enables the pairing service and restarts the Bluetooth Support Service.

Use when: a device will not pair or connect.

D System_File_Scan_Repair.bat

- DISM first, then SFC, the order Microsoft recommends (15-40 minutes). ResetBase is an optional Y/N question because it cannot be undone.

Use when: programs crash, freeze often, or you suspect damaged system files.

External tools (optional)

Not part of this repository. Use only if you want them

Read before use

Both tools download and run the latest code from the internet each time. What was checked in October 2026 may differ from a future version. Create a restore point first (menu 6) and run 3_Check_Status afterwards to see which of this project's values changed.

Tool	FPS effect	Real benefit	Risk	For
WinUtil: Install tab	None	Installs several programs at once through winget	Low	Everyone
WinUtil: Tweaks tab	Almost none	Overlaps almost entirely with this project	Medium: may change values set here	Not recommended after this project
RemoveWindowsAI	Very little to none	Removes Copilot and background AI processes; saves some RAM and disk; privacy	High: force-removes packages Windows protects	People who never use AI, or PCs with 8 GB RAM or less

W Chris Titus Tech WinUtil

- Use the Install tab. Skip the Tweaks tab; if you used it, run 3_Check_Status and re-apply what changed.
- To open: right-click Start > Terminal (Admin), then paste the command from the official GitHub page (Stable):

```
irm https://christitus.com/win | iex
```

Link: github.com/ChrisTitusTech/winutil

R RemoveWindowsAI (zoicware)

- Removes Copilot, Recall and other AI features, and installs packages so Windows Update does not bring them back.
- If you only want Copilot to stop using RAM, you do not need this: turn it off in Settings > Apps > Startup, or uninstall the Copilot app in Settings > Apps > Installed apps.
- Needs **Windows PowerShell 5.1** (search "Windows PowerShell", run as administrator). Its GitHub page says PowerShell 7 has problems.
- Always use Backup Mode. It creates a restore point and is what makes a full revert possible.
- The GitHub page says some antivirus software may complain and suggests turning antivirus off temporarily. **This manual advises against that.** If it is blocked, do not use it.
- Read the Documentation page first: the latest version has options touching Windows Defender's AI components.

Backup Mode (removes everything):

```
& ([scriptblock]::Create((irm "https://raw.githubusercontent.com/zoicware/RemoveWindowsAI/main/RemoveWindowsAi.ps1"))) -nonInteractive -backupMode -AllOptions
```

Revert everything:

```
& ([scriptblock]::Create((irm "https://raw.githubusercontent.com/zoicware/RemoveWindowsAI/main/RemoveWindowsAi.ps1"))) -nonInteractive -revertMode -AllOptions
```

Each command must be on one line. If copying from this PDF splits it, join it before pressing Enter, or copy from the GitHub page.

Link: github.com/zoicware/RemoveWindowsAI. Use the full links above, never shortened ones, because you cannot tell where a short link leads.

References

Links are clickable in the PDF. Last checked 30 Sep 2026

Used for	Source / link
03 SystemResponsiveness, Tasks\Games	Microsoft Learn - Multimedia Class Scheduler Service learn.microsoft.com/en-us/windows/win32/procthread/multimedia-class-scheduler-service Values below 10 are raised to 20; a value of 100 turns MMCSS off.
04 Win32PrioritySeparation	Microsoft TechNet - Win32PrioritySeparation technet.microsoft.com/library/Cc976120 Default is 0x2, as in UNDO_04.
07 Delivery Optimization	Microsoft Learn - Delivery Optimization reference learn.microsoft.com/en-us/windows/deployment/do/waas-delivery-optimization-reference
08 GPU scheduling	Microsoft DirectX Blog - Hardware Accelerated GPU Scheduling devblogs.microsoft.com/directx/hardware-accelerated-gpu-scheduling/ Most users should not expect a big change.
13 Memory Integrity	Microsoft Support - Options to optimize gaming performance in Windows 11 support.microsoft.com/en-us/windows/options-to-optimize-gaming-performance-in-windows-11-a255f612-2949-4373-a566-ff6f3f474613 Confirms an effect on some PCs and warns of higher exposure to threats.
Menu 3: TCP per adapter	Microsoft Support KB328890 - TcpAckFrequency support.microsoft.com/kb/328890 Set per interface GUID; Microsoft advises against changing defaults without testing.
Menu 4, power plans	Microsoft Learn - Powercfg command-line options learn.microsoft.com/windows-hardware/design/device-experiences/powercfg-command-line-options

Menus 5, 6: restore point	Microsoft Learn - Checkpoint-Computer learn.microsoft.com/powershell/module/microsoft.powershell.management/checkpoint-computer Normally one restore point per day, so the script sets SystemRestorePointCreationFrequency = 0 first.
15 Timer resolution	Microsoft Learn - timeBeginPeriod function learn.microsoft.com/en-us/windows/win32/api/timeapi/nf-timeapi-timebeginperiod Describes the per-process change since Windows 10 2004 and Windows 11, and the extra power use of a higher resolution.
16 CPU boost	Microsoft Learn - PERFBOOSTMODE and processor power management options learn.microsoft.com/en-us/windows-hardware/customize/power-settings/options-for-perf-state-engine-perfboostmode learn.microsoft.com/en-us/windows-hardware/customize/power-settings/configure-processor-power-management-options Boost mode 2 = Aggressive. Energy performance preference runs 0-100 %.
WinUtil	GitHub - ChrisTitusTech/winutil (README) github.com/ChrisTitusTech/winutil
RemoveWindowsAI	GitHub - zoicware/RemoveWindowsAI (README) github.com/zoicware/RemoveWindowsAI
repair-tools: SFC / DISM	Microsoft Support - Use the System File Checker tool support.microsoft.com/help/929833 Run DISM before SFC, the order used here.

Without official documentation

- 01 GameBar/DVR and 09 Fullscreen Exclusive (GameDVR_FSE* values): shared by the tweaking community; no Microsoft page explains them.
- 02 PowerThrottlingOff, EnergyEstimationEnabled: no public Microsoft documentation.
- 15 GlobalTimerResolutionRequests: reported by the community. The Microsoft timeBeginPeriod page describes the behaviour change but not this value.
- 06 Control Panel\Desktop timeouts, MinAnimate: widely known, not sourced here.
- 11 Keyboard Response flags and 12 Ndu memory leak: from user reports, not Microsoft.
- The defaults in reg_undo for these are typical Windows values. If you need your exact originals, use the Backup folder.

Results differ between PCs and Microsoft may change how these values behave in a later Windows release. Every item can be reverted with menu 7 or the Backup folder.